

Practical 5: Mixed-Effect Models

First author name goes here

You can have more than one author

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Instructions

Your goal is to practice fitting a mixed-effects model (a model with one or more random effects).

Choose a Dataset

To begin, consider the datasets you've worked on so far in earlier sessions. Which one of them might you want to try adding random effect(s) to? It might be one where...

- Your fitted model failed assessment because of a problem with residual independence; or,
- You noticed a predictor variable that is a good candidate for a random effect.

If you are bold and ambitious, you might instead consider trying to fit a model (with at least one random effect) to a dataset of your own or a publicly available dataset you work with (that you have easy access to).

If you have time, you can try more than one option!

If you want a fresh dataset to try, we have those too (just ask).

And bear in mind: not *every* regression model *needs* to be a mixed-effects model (so if you really think there aren't any random effect variable candidates in your dataset, you may be right!).

Tasks

Plan, fit, assess and interpret a revised model with at least one random intercept included.

What do you notice? Any differences from before?

If you decide to show prediction plots from the model, consider talking with an instructor about interpretation (which is subtly different).